

ABEL SANTAMARIA, CubaWMO: **783465**Lat: **22.492N**Lon: **79.944W**Elev: **338**StdP: **14.52**Time Zone: **-5.00 (CUB)**Period: **05-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.0	53.5	46.0	46.5	56.1	48.2	50.6	59.9	21.6	75.5	20.0	76.0	2.2	270	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	91.8	74.9	91.4	75.0	90.0	75.1	78.7	86.1	78.1	85.9	77.4	85.7	9.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.8	141.4	82.8	75.5	135.3	81.6	75.2	134.1	81.3	42.2	86.1	41.6	86.2	41.0	86.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.4	16.4	44.4	94.7	3.1	1.1	42.2	95.5	40.4	96.1	38.7	96.7	36.5	97.5		
(5)			43.6	81.1	3.4	2.0	41.1	82.5	39.2	83.6	37.3	84.8	34.8	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.2	70.1	71.1	72.1	76.0	78.0	80.3	81.3	81.4	80.4	78.2	73.1	72.2		
	DBStd	5.45	5.02	5.19	5.13	3.46	2.71	1.88	1.48	1.46	1.28	2.75	3.89	5.09		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	54	17	14	10	0	0	0	0	0	0	0	1	12		
	CDD50	9571	624	590	685	781	869	910	970	974	913	874	692	689		
	CDD65	4150	176	184	230	331	404	460	505	509	463	409	243	236		
	CDH74	33518	1025	1277	1842	2775	3365	4078	4821	4664	3855	2966	1464	1386		
(13)	CDH80	12509	213	336	580	1038	1321	1678	2143	2069	1572	975	278	306		
(14)	Wind	WSAvg	6.7	6.8	7.5	8.0	8.0	7.2	6.3	6.0	5.1	6.0	6.5	6.4		
(15) Precipitation	PrecAvg	50.7	1.4	1.3	1.5	2.4	6.4	7.0	5.8	6.6	7.5	6.3	2.6	1.7		
	PrecMax	64.2	3.0	4.6	3.3	4.6	12.9	14.3	8.5	9.8	13.7	18.9	7.1	8.0		
	PrecMin	33.1	0.3	0.3	0.6	0.8	2.4	3.3	3.4	4.3	4.4	2.0	0.3	0.4		
	PrecStd	6.9	0.7	0.9	0.6	0.8	3.0	2.6	1.5	1.3	1.9	3.4	1.8	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.9	88.1	90.0	91.8	93.2	92.8	93.5	93.4	91.8	90.0	87.7	87.9		
		MCWB	69.5	69.3	69.8	70.9	72.1	75.4	75.0	75.7	75.8	75.8	74.7	72.3		
	2%	DB	84.4	86.1	88.0	89.9	91.1	91.2	91.7	91.7	90.4	88.6	85.0	85.7		
		MCWB	69.9	69.3	69.2	70.8	72.5	75.6	75.6	75.9	76.0	76.0	73.8	72.6		
	5%	DB	82.5	84.2	86.0	88.1	88.2	89.7	91.1	91.2	89.5	87.6	82.8	83.8		
		MCWB	70.1	69.3	69.6	70.6	72.6	75.9	75.8	75.9	76.2	76.0	72.8	72.0		
	10%	DB	80.4	82.2	83.8	86.2	87.4	88.1	89.5	89.6	87.9	85.8	80.9	81.0		
		MCWB	69.6	69.4	68.9	70.6	72.5	75.8	76.0	76.2	76.5	75.7	71.5	71.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	73.7	73.8	75.3	76.9	79.0	79.0	79.5	79.2	78.9	77.0	75.8		
		MCDB	80.0	80.9	82.8	84.4	85.6	86.2	87.2	86.8	85.4	84.9	82.7	81.7		
	2%	WB	72.5	72.3	72.6	73.9	75.6	77.9	78.1	78.6	78.3	77.7	75.7	74.5		
		MCDB	78.9	79.3	81.3	82.9	84.2	85.8	86.6	86.7	85.2	84.5	81.1	80.7		
	5%	WB	71.6	71.3	71.5	72.9	74.7	77.0	77.4	77.8	77.6	76.9	74.7	73.6		
		MCDB	78.0	78.4	80.2	81.9	83.2	85.8	86.2	86.4	85.3	84.1	80.0	79.8		
	10%	WB	70.7	70.6	70.5	72.2	74.1	76.3	76.8	77.1	77.0	76.2	74.0	72.7		
		MCDB	77.1	77.8	79.1	81.0	82.5	84.9	85.7	86.1	85.3	83.0	79.1	78.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.5	21.6	21.3	19.1	17.1	18.3	18.1	17.1	15.0	16.7	17.9		
		MCDBR	20.9	22.4	23.3	23.2	21.3	18.7	19.7	19.7	18.7	17.4	17.7	19.1		
		MCWBR	9.2	9.3	8.6	7.9	7.5	7.1	7.1	7.3	7.4	7.3	8.7	8.5		
	5% WB	MCDBR	17.8	19.2	20.8	20.5	18.0	17.4	18.0	17.9	17.0	15.3	14.1	16.0		
		MCWBR	8.3	8.5	8.5	7.5	7.1	7.3	7.4	7.6	7.6	7.1	7.6	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.355	0.360	0.376	0.397	0.423	0.458	0.464	0.441	0.416	0.404	0.376	0.377		
	taud		2.528	2.516	2.460	2.395	2.368	2.301	2.272	2.355	2.434	2.454	2.486	2.475		
	Ebn at Noon		290	295	294	287	277	266	264	271	278	277	280	277		
	Edh at Noon		30	32	36	39	40	42	43	40	36	34	31	31		
(44) All-Sky Solar Radiation	RadAvg		1252	1531	1788	1964	1897	1805	1974	1893	1696	1453	1290	1170		
	RadStd		90	52	74	64	190	133	109	70	78	101	70	95		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page